

Project Overview

Title: Library Management System

Problem

Many libraries still rely on manual processes for cataloguing books, tracking inventory, and managing borrowing records.

Solution

Developed a Python and MySQL-based Library Management System that automates book management, user management, issue/return tracking and personal note-taking.

Key Features

- Administrator Functions
 - Administrator login
 - Add, update, delete and search books
 - Add, update, delete and search users
 - View all books and users
 - Issue books to members
 - Process book returns
- Member Functions
 - Member login
 - Search books
 - View available books
 - Borrow books
 - Return books
 - View borrowing history
 - Create, update, delete and search personal reading notes

Technologies Used

- Python
- MySQL
- MySQL Connector
- SQL

Database Design

The system uses a relational database consisting of the following entities:

Table	Purpose
Admin	Stores administrator credentials
Users	Stores member information
Books	Stores book inventory details
Issued	Tracks issued and returned books
Notes	Stores personal reading notes

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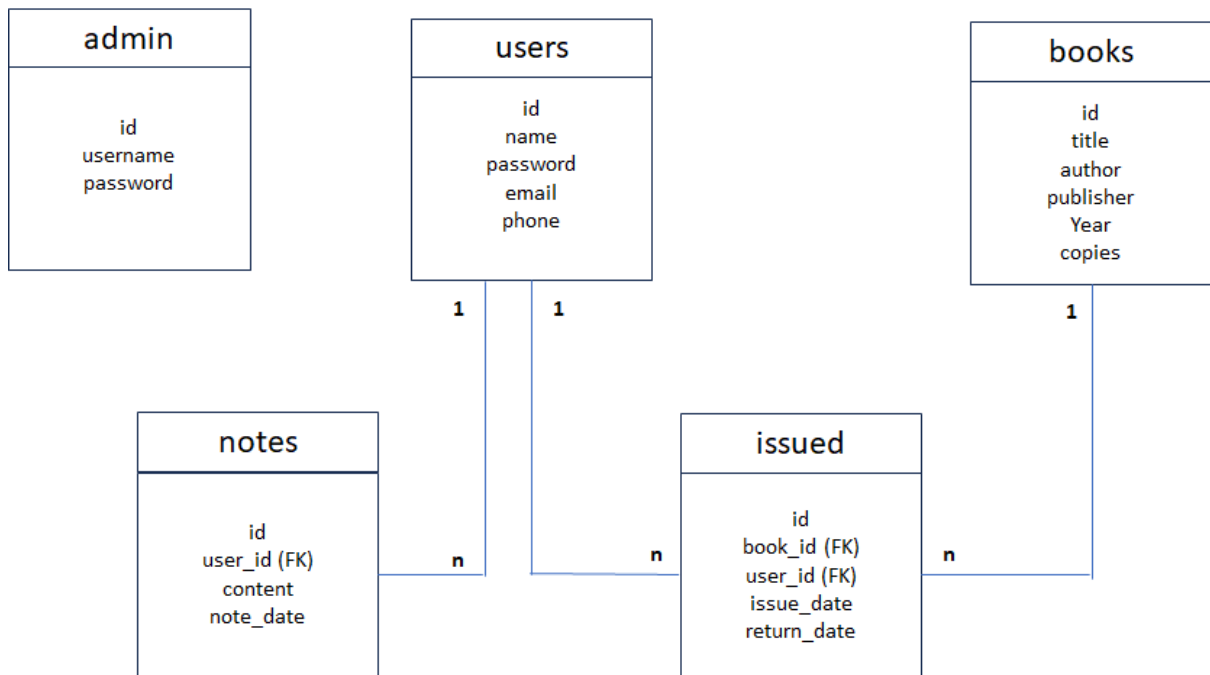


Fig-1: Database Entity Relationship

Key Learning Outcomes

- Database design
- Secure coding practices
- Software development lifecycle
- Problem-solving through technology

Future Enhancements

- Graphical user interface
- Password encryption
- Format validation (Email, Telephone, Year etc.)
- Barcode integration
- Recommendation engine

The project helped me understand how systems are designed. I realised that technology is not just about writing code. It is about understanding how people work, how information is organised and how systems can be designed to make processes more efficient. Working on this project strengthened my interest in exploring the connection between technology, society, and problem-solving.